

Capstone Project: Milestone Three

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Project: Financial Risk Modeling: From Loan Approval to Default and Bankruptcy

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1. Description of the capstone project

This capstone project focuses on financial risk modeling in the finance industry, examining the risk assessment process from loan approval to default and bankruptcy. Financial institutions play a crucial role in the stability of the global economy, and many major financial crises have been associated with weaknesses in credit assessment, excessive risk-taking, and poor monitoring of financial obligations. For this reason, banks and other financial institutions must carefully evaluate borrowers and counterparties before entering into financial relationships.

The project uses real-world datasets to analyze different stages of financial risk. The Modified Loan/Application Register dataset represents the initial decision-making stage, where a financial institution decides whether to approve or deny a loan application. The Default of Credit Card Clients dataset captures the behavioral stage, focusing on repayment patterns and characteristics associated with future default. Finally, the Polish Bankruptcy dataset addresses the most severe outcome: whether a company is likely to fail financially and become unable to meet its obligations.

Together, these datasets support a multi-stage predictive framework for financial risk modeling. The framework connects loan approval decisions, borrower repayment behavior, and bankruptcy prediction into one broader analytical perspective. The goal of the project is to identify key financial, demographic, and behavioral features that may help predict risky decisions, support credit assessment, improve portfolio allocation, and reduce potential losses.

The combined insights from these analyses can support financial institutions in making more informed decisions, detecting high-risk clients earlier, and implementing proactive risk management strategies.

2. Polish Bankruptcy Prediction dataset (Tomczak, 2016): 43,405 rows and 66 columns

2.1. The Polish Bankruptcy Prediction dataset was collected from the Emerging Markets Information Service (EMIS), a database that provides information on emerging markets worldwide, for the period 2007 to 2013 for bankrupt companies and 2000 to 2012 for still operating companies. According to the data source and Tomczak et al. (2016), who analyzed this dataset, it was decided to focus on financial data from the manufacturing sector. In Poland, since 2004, many companies in this sector have gone bankrupt, and it was possible to obtain representative samples. The dataset is publicly available through the UC Irvine Machine Learning Repository.

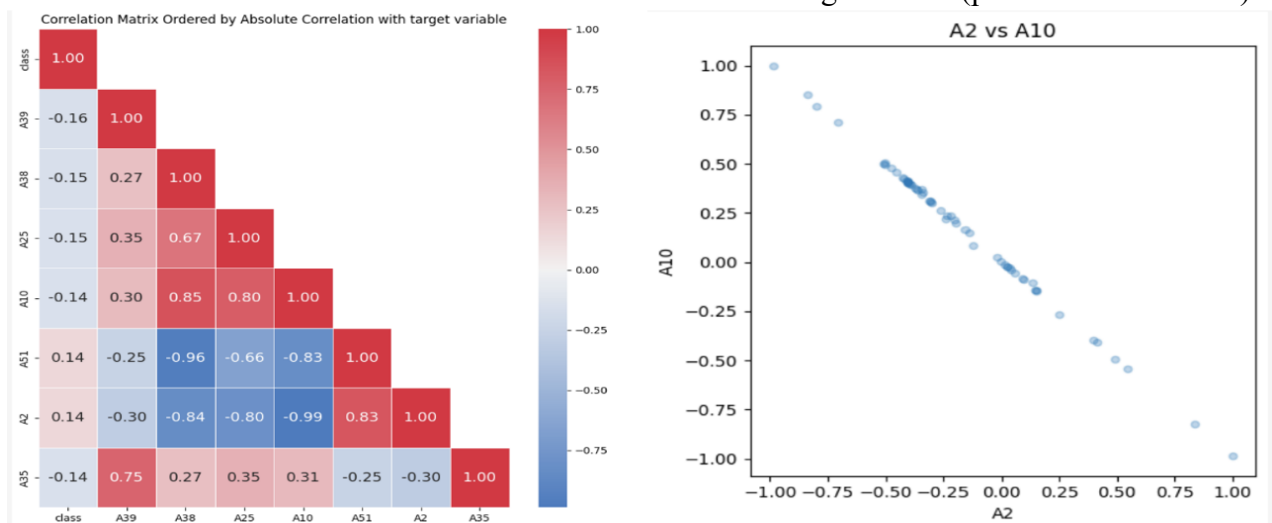
It contains **43,405 firm-year observations (rows) combined and 66 variables** (64 float, 2 integers), primarily financial ratios and performance indicators. It includes a binary target variable (Bankrupt?). It is divided into 5 forecasting periods, e.g., 1st Year – the data contains financial rates from the 1st year of the forecasting period and corresponding class label that indicates bankruptcy status after 5 years, 2nd Year – the data contains financial rates from the 2nd year of the forecasting period and corresponding class label that indicates bankruptcy status after 4 years, and so on. Each of these classification cases has a different set of companies. The features are well described on the source website and can be grouped into economically meaningful categories, including **profitability measures** (e.g., gross profit on sales), **leverage and solvency ratios** (e.g., total liabilities / total assets), **liquidity** (e.g., current assets / short-term liabilities), **time-based stress indicators** and **efficiency metrics** (e.g., sales / inventory). Those features are financially meaningful and Interpretable, but some of them may be highly redundant and correlated. Also, after initial exploration it was discovered that because of ratio-based nature some values are unstable.

An exploratory analysis was conducted to assess the quality of variables. Initially, this dataset contained 400 duplicates ($\sim 0.9\%$), which needed to be removed. Also, 23,438 rows had at least one missing value. The overall ratio of missing values was $\sim 1.5\%$. It was discovered that 5 columns, including quick assets to long term liabilities and sales-growth-ratio, are mostly the main reason for this high percentage. After the diligent assessment, it was decided that these columns can be safely removed. The final missing values were 2887 rows with at least one missing value, for a general rate of $\sim 0.35\%$, which is a very satisfactory result. After initial data cleaning 2,083 of 40,921 firms ($\sim 5.1\%$) are bankrupt. To address this issue, techniques such as oversampling (e.g., SMOTE), undersampling, or class-weight adjustment, that is higher penalty to mistakes made on the minority class, may be applied (Wang & Liu, 2021). Additionally, multicollinearity among financial ratios is expected, and Pearson correlation analysis can help in identifying and reducing redundant features.

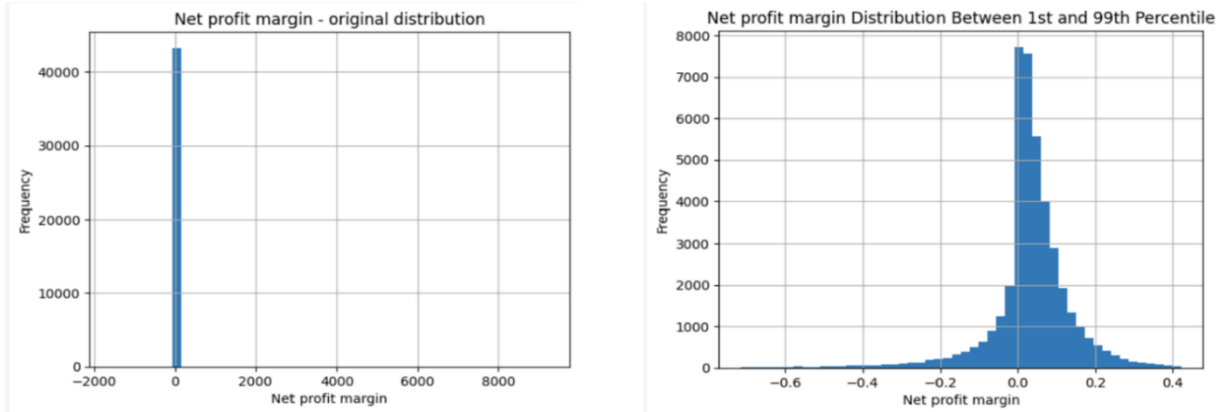
2.2. There are also extreme values because the features are financial ratios, and ratios can explode when the denominator is small or unstable; e.g., companies near bankruptcy can have a profit of ~ 0 and unstable liabilities, or negative cash flow. Also, it can be caused by accounting quirks, such as depreciation adjustments, one-time expenses, and extraordinary items. Extreme values in financial data may signal genuine distress rather than noise. Instead of removing them entirely, use techniques such as winsorization combined with indicator variables (flagging) to preserve their informational value while maintaining model stability. Winsorization can “clip” extreme values to the 1st and 99th percentile. Only highly skewed variables should be “clipped” ($\text{skew} > 2$) to preserve meaningful variation in stable features. The last steps in handling outliers should be to transform highly skewed distributions (e.g., log / signed log) and scale the variables (e.g., StandardScaler). The presence of outliers requires a proper and diligent approach during the

modeling phase. Log transformation should be performed with care when the values are naturally bounded (e.g., ROA /ROE) or are less than -1. A better approach is to keep the original scale, use robust scaling, or apply a signed log transformation (which preserves the direction). Taking everything into consideration, the visualization of the distribution of the skewed features between the 1st and 99th percentiles showed that, without extreme outliers, our data takes on a more stable shape, which can be further transformed for better predictions.

2.3. An additional advantage of winsorization is more stable and stronger Pearson correlation scores. Before clipping values, extreme outliers distorted the scores, introducing noise that biased the results, and the highest correlation with the target was approximately 5%. After clipping values between the 1st and 99th percentile, the Pearson scores improved. However, the Pearson correlation measures only the linear relationship between a single ratio and bankruptcy. Bankruptcy is more likely to depend on combinations such as high leverage, weak liquidity, and poor cash flow. There are some highly correlated values, e.g., A10 (equity / total assets) and A2 (total liabilities / total assets), which is normal for ratios derived from nearly identical financial indicators. We should remove highly correlated values to avoid multicollinearity and unstable coefficient estimates. The most correlated feature with the target is A39 (profit on sales / sales).



Figures 1 and 2: Polish Bankruptcy dataset: Correlation Matrix with the target (to the left) and negative multicollinearity -0.99 to the right.



Figures 3 and 4: Net profit margin - original distribution with extreme outliers (to the left) and winsorized Net profit margin distribution between the 1st and 99th percentile (to the right).

3. Modified Loan/Application Register (Consumer Financial Protection Bureau, 2025)

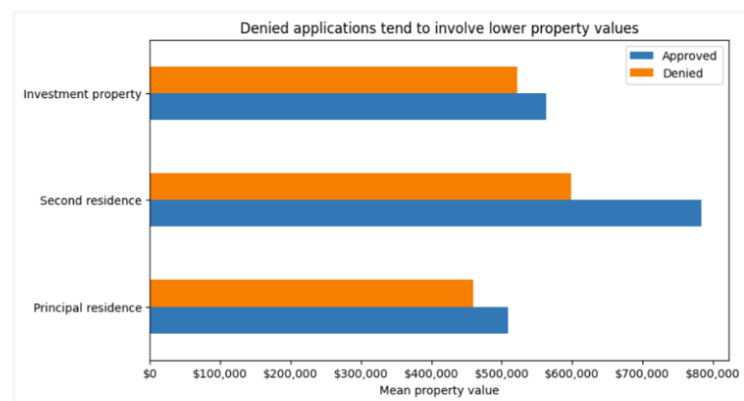
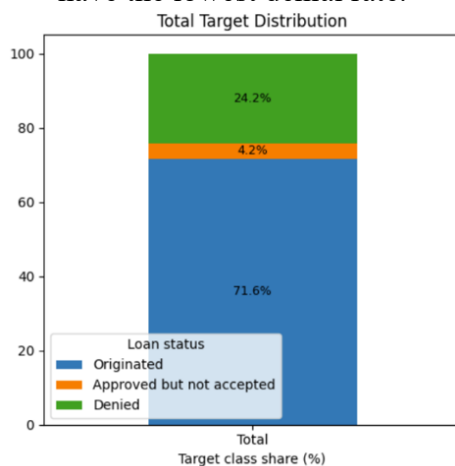
3.1. The Modified Loan/Application Register (LAR) dataset is published annually under the Home Mortgage Disclosure Act (HMDA). It provides detailed loan-level information on mortgage applications submitted to financial institutions across the US. The 2024 dataset includes millions of observations, and both approved and denied mortgage applications, making it suitable for financial risk modeling in credit decision-making. A data dictionary and supporting documentation accompany the dataset. Each column in the dataset is reported using coded values. For example, the variable `action_taken` does not store “Approved” or “Denied” as text; instead, it stores a numerical value. To start analyzing this dataset, I had to download the raw data file from the official website of the United States government. The entire file was about 3 GB; thus, I had to reduce it to a size more suitable for the computer’s RAM to continue the analysis. I divided this dataset into 100,000-row chunks and sampled 20% of the observations from each chunk to avoid structural patterns and retain representative samples. At the end of this process, I finalized the dataset, which contained 2,451,832 observations and 85 features. I saved the file in the columnar binary storage format (parquet), which is more suitable for big datasets, *faster*, and *retains the*

data types. I excluded columns with many missing values and selected important features, including target variable, location, loan/application characteristics, borrower financial variables, and applicant demographics what gave me 57 features in total. I also changed original object type assigned to all the columns for the proper 10 numeric, 43 category, and 4 string types. In addition, after analysis of the values distribution, I excluded the invalid and economically impossible values from the numeric columns (eg., income below 0 and above 10,000,000 \$) and prepared the final target variable by filtering dataset only on 3 values: 1: Originated, 2: Approved but not accepted, 3: Denied, and then in the modelling step combining class 1 with 2 two obtain binary target. The final dataset has 630 duplicates (0.036 %), The total proportion of missing values is 5%, but this result mostly accounts for the 5 columns with the highest proportion of missing values (co_applicant_age_above_62: 59%, co_applicant_age: 59%, total_loan_costs: 49%, origination_charges: 49%, interest_rate: 27%), which will be used only for visualization and dropped for the final modelling phase, because of the leakage problem. The target variable is moderately imbalanced, which is natural for loan approval and acceptance. We can implement various techniques to mitigate this problem, e.g, SMOTE. About 76% is for the combined approved class, and 24% is for the denied class.

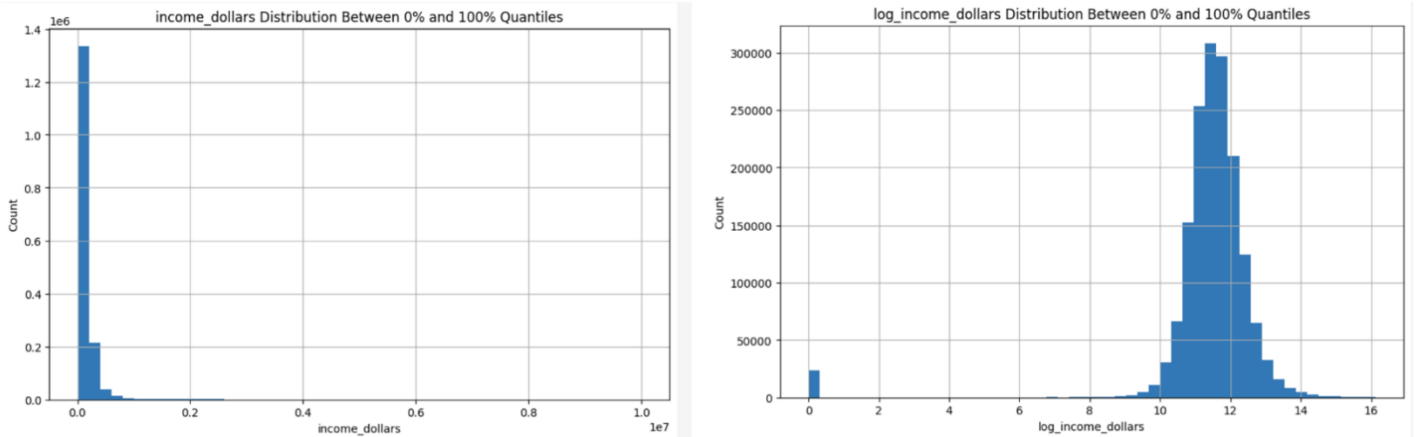
3.2. The numerical features of the LAR dataset have many outliers (even after removing the economically impossible values) and are highly skewed. To present the problem I provided two plots that show gradual improvements in representing the distribution of the income in dollars feature, which will also stabilize feature model built based on these data: 1) original distribution, 2) log transformed (log1p) variable. The log transformation substantially improves the shape of loan_amount and property_value, which now look much closer to log-normal distributions. The original loan_amount skewness and kurtosis were ~6.04 and ~81, respectively, and its log

skewness and kurtosis were ~ -0.5 and ~ 0.47 , respectively. Also, the original `property_value` skewness and kurtosis were ~ 5.7 and ~ 58.7 , respectively, and its log skewness and kurtosis were ~ 0.02 and ~ 1.3 , respectively. However, the transformation is less appropriate for `combined_loan_to_value_ratio`, where skewness and kurtosis worsen. Although the log transformation substantially compresses the income distribution, a visible spike at zero remains because $\log_1 p(0) = 0$. This creates a separate mass far from the main log-income distribution and explains the strong negative skewness after transformation.

3.3. The next phase of the analysis was bivariate analysis. I selected numeric features to create a Pearson correlation matrix and ordered them by their absolute correlation with the target variable. Then, to avoid clustering the plot, I created a mask that shows only the lower triangular part of the matrix. The most correlated variables are `loan_amount`, `interest_rate`, `has_co_applicant`, `combine_loan_to_value_ratio`, `income`, `loan_terms`, and `property_value`. I can see that the features `income` and `income_dollars` are highly redundant because their correlation is 100%. It makes complete sense. Income dollars is just income multiplied by 1,000. Furthermore, I observed that denial applications tend to involve lower property value and that investment property applications have the lowest denial rate.



Figures 5 and 6: Target distribution in the LAR dataset (to the left) and target distribution across the property value (to the right).



Figures 7 and 8: LAR: Original Income dollar distribution (to the left) and log-transformed income distribution with a visible spike at 0 (to the right).

4. Default of Credit Card Clients (Yeh, 2009): 30,000 rows and 24 columns

4.1. The dataset was collected by Yeh (2009) to conduct research on credit card default behavior in Taiwan and is based on the payment data from October, 2005, collected from a major bank in Taiwan. During the years 2005 - 2006, banks in Taiwan overissued cash and credit cards to applicants who were not sufficiently qualified, and many cardholders overused credit cards for consumption. As stated in the original study, the objective is to “predict business performance or individual customers’ credit risk and to reduce the damage and uncertainty”(Yeh & Lien, 2009).

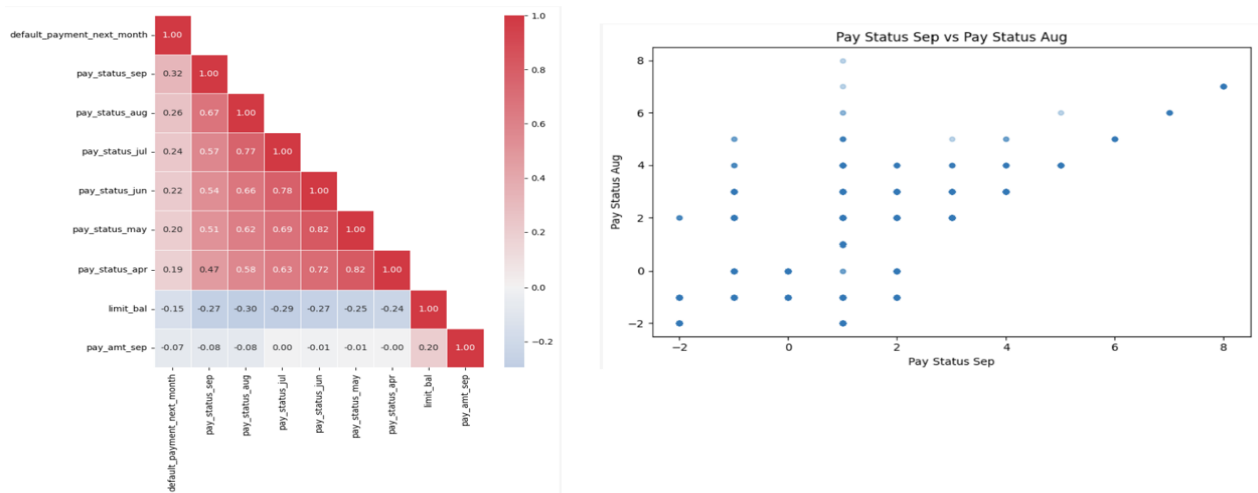
The dataset consists of 30,000 non-null observations and is unbalanced, with approximately 22.12% of cardholders defaulting. The financial features may also introduce multicollinearity. Furthermore, this dataset includes 23 explanatory variables and one binary dependent variable (whether a client defaulted on their credit card payment). The feature meanings are described in Yeh and Lien (2009). Although all features are stored as integer values, they represent a mixture of continuous financial measures (e.g., credit limits, bill amounts, payment amounts), ordinal repayment-status variables, and categorical demographic characteristics (e.g., age, education). This numerical encoding allows direct application of machine learning algorithms. Treatment of categorical and ordinal variables with appropriate techniques, such as one-hot encoding or ordinal

encoding, may be necessary to prevent the model from learning artificial numerical ordering and continuous relationships. Some categorical variables (MARRIAGE, EDUCATION) contain undocumented or special categories, such as education codes 0, 5, and 6 and marriage code 0. I would keep them as separate categories or group them into Other/Unknown before modeling. This dataset contains only 35 duplicate rows; thus, it is safe to remove them.

4.2. Several monetary features, including X12–X17 (bill statements) and X18–X23 (repayment amounts), display significant positive skewness (skewness > 2), indicating heavy-tailed distributions. This behavior is typical in financial datasets, where most observations are relatively small, but a minority of cases involve very large transaction values. Although these observations are valid, such distributions can distort model training by introducing scale imbalances and increasing sensitivity to outliers. To mitigate these effects, transformations such as the log or signed log are recommended to normalize the distribution and improve numerical stability. There is a clear difference in a visual presentation of the distribution when the focus is only on data between 1st and 99th percentile, which supports further useful insights about the data.

4.3. Historical payment status is the dominant predictor group, but the six monthly status variables contain considerable overlapping information. The correlation among features in this group makes intuitive sense, because a customer who was recently late with repayments is more likely to default in the following month. Furthermore, the most recent month, September, has the strongest correlation. The relationship becomes progressively weaker for later months, suggesting a recency effect: recent repayment behavior contains more relevant information about the upcoming default. We also observe that the payment-status variables are strongly correlated, indicating that customers' repayment behavior tends to persist over time. Someone late in one month is likely to remain late in another month. This creates substantial multicollinearity. In my opinion, we should

not exclude them from the model in order to retain useful temporal information. `limit_bal` appears useful. The negative sign suggests that, on average, customers with higher credit limits have a lower probability of default. That does not strictly mean that simply increasing someone's credit limit prevents default. Higher credit limits may be given to customers who already have higher income, stronger credit histories, more stable employment, and, in general, lower estimated credit risk.



Figures 9 and 10: Correlation matrix features with the target variable in the Credit Card dataset (to the left) and the positive relationship of the `pay_status` features (to the right)

5. Conclusion

5.1. Several aspects of the exploratory data analysis were expected. First, all three datasets showed characteristics commonly observed in financial risk data: class imbalance, skewed numerical distributions, extreme values, and multicollinearity. In the Polish Bankruptcy dataset, the low proportion of bankrupt firms was expected because bankruptcy is a relatively rare event (similarly the Default of Credit Card Clients target). The LAR dataset also showed an imbalanced but usable target distribution, with approved applications forming the majority class and denied applications forming the minority class. It was also expected that financial variables would contain heavy-tailed distributions, and the presence of outliers did not automatically indicate poor data quality. The

bivariate analysis also confirmed several expected relationships. In the Polish Bankruptcy dataset, financial ratios based on similar accounting components were highly correlated, which is normal when variables are derived from related balance-sheet or income-statement values. In the Default of Credit Card Clients dataset, recent repayment status was more strongly associated with default than older repayment status, suggesting a recency effect. This is financially intuitive because a customer who is currently behind on payments is more likely to default in the near future.

However, some findings were unexpected or required more careful interpretation. In the LAR dataset, log transformation improved the distributions of `loan_amount` and `property_value`, but it was less appropriate for `combined_loan_to_value_ratio`. This shows that transformation should not be applied mechanically to every skewed variable. Another unexpected issue was the spike at zero in transformed income, because $\log(0)$ remains zero and creates a separate mass away from the main income distribution. This suggests that zero or missing-like income values may need separate treatment before modeling. Another unexpected finding was that some variables with strong practical importance may create leakage problems. For example, LAR variables such as interest rate, total loan costs, and origination charges may be useful for visualization, but they should be removed from the final supervised model if they are only known after approval or are directly connected to the decision outcome. A model may appear accurate while actually learning information that would not be available at the time of prediction.

Overall, the expected findings confirmed that the datasets are suitable for financial risk modeling, while the unexpected findings showed that careful preprocessing and feature selection are essential. The data should not be treated as clean, normally distributed input for a model. Instead, the exploratory analysis suggests that reliable modeling will require thoughtful handling of outliers, skewness, missing values, leakage, class imbalance, and redundant features

5.2. These datasets should not be disqualified. All of them contain a suitable number of observations for machine learning and have clearly defined target variables. Although the datasets contain extreme outliers and highly skewed features, these issues are common in financial data and do not automatically make the datasets unsuitable for analysis. Techniques such as winsorization, transformation, and robust scaling can reduce the influence of extreme values and make the central part of the distribution easier to interpret. Furthermore, multicollinearity is another common characteristic of financial datasets, especially when many variables are derived from related financial ratios or borrower characteristics. Pearson correlation analysis can help identify highly correlated feature pairs and support the decision about which redundant features should be retained or removed before further modeling.

5.3. Taking into account the different bankruptcy forecasting horizons, the Polish Bankruptcy dataset may support not only binary supervised classification but also a multiclass classification. In such a setup, a multiclass model with a softmax output layer could be used to predict one of six classes: five classes representing different time horizons to bankruptcy and one class representing non-bankrupt firms. Unsupervised techniques (after excluding target variable) such as clustering and anomaly detection can also be applied to discover hidden patterns in data and new features.

The Default of Credit Card Clients and LAR datasets are well suited for supervised binary classification because both contain, or can be transformed to contain, binary target variables. Gradient boosting models such as XGBoost, CatBoost, and LightGBM are strong options because they handle mixed financial, demographic, and categorical features effectively. Unsupervised methods such as K-Means may also support exploratory segmentation by identifying lower- and higher-risk groups based on repayment behavior, credit limits, income, DTI, LTV, loan characteristics, and demographic variables.

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